Post-GWAS Analysis

Gene and Functional Annotation

Training Material
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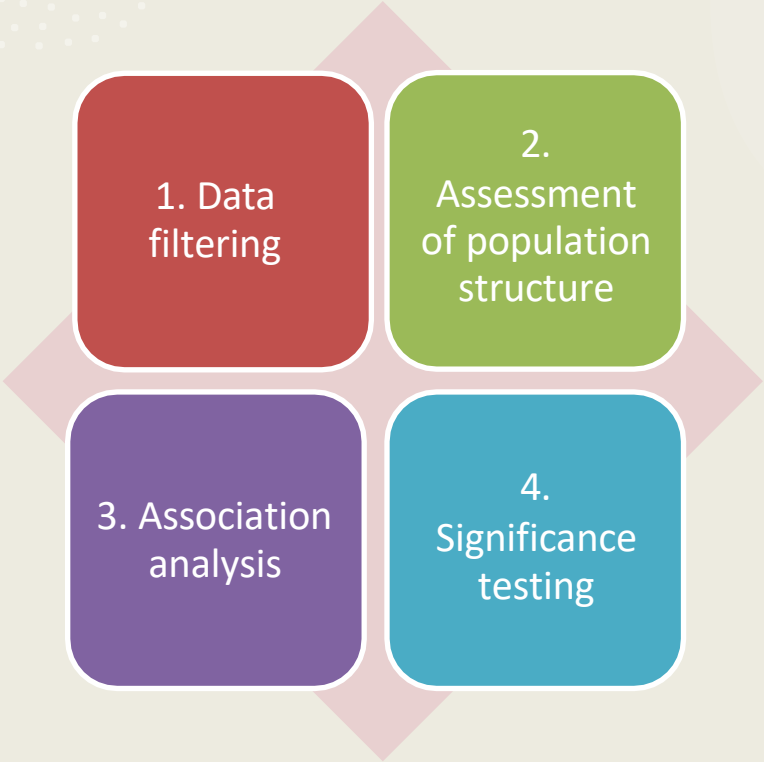
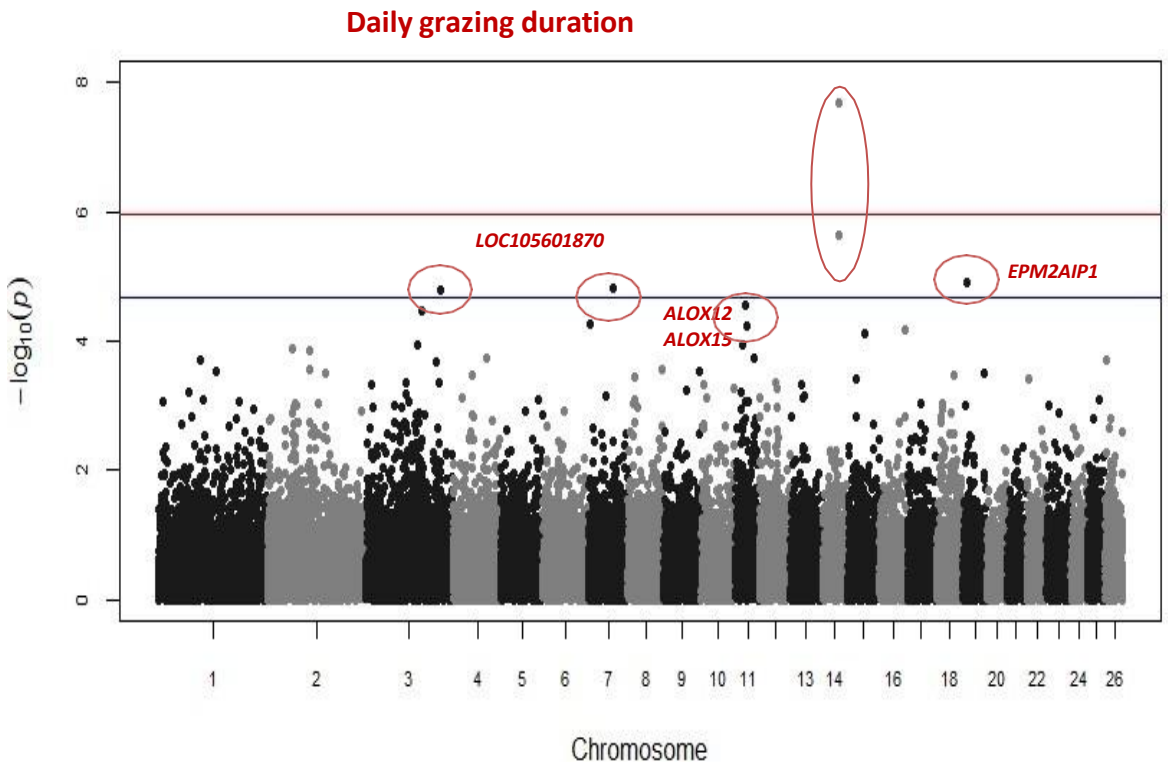
Summarize GWAS steps - 1st Lecture



GWAS results – genome-wide level

Manhattan plot

Association between SNPs and phenotype under study



Chromosomes of interest – 3, 7, 11, 14, 19

SNPs identified on chr3 will be used for the functional annotation

Post-GWAS analysis - SNP and Gene Annotation

Detection of candidate genes up/down-stream of the identified genomic markers **within a defined distance** (bp)

Why is this process needed?

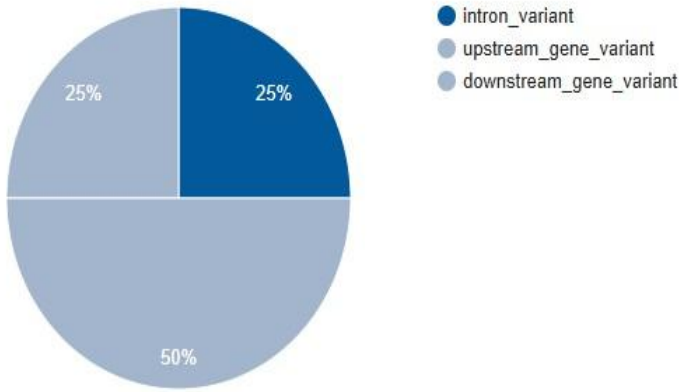
Post-GWAS analysis - SNP and Gene Annotation

We link variants to nearby genes and their functions to biologically interpret GWAS results and prioritize the genomic markers

- **Genomic Context**
  - Marker position: exonic, intronic intergenic, UTR, nearest gene or regulatory element

- **Predicted Consequence of variant change**
  - synonymous, missense, nonsense mutations

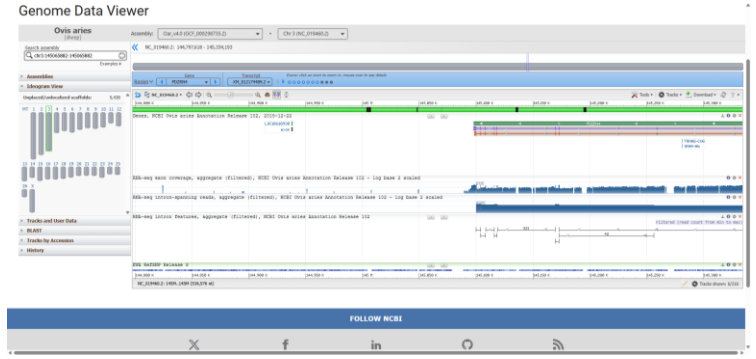
Consequences (most severe)



Consequence type	Count
intron_variant	1
upstream_gene_variant	2
downstream_gene_variant	1

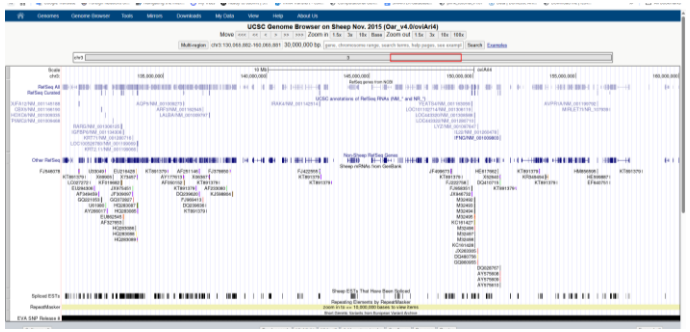
Genome Browsers to explore genomic regions closely located to the identified markers

NCBI Genome Data Viewer



https://www.ncbi.nlm.nih.gov/gdv/browser/genome/?id=GCF_000298735.2

UCSC Genome Browser



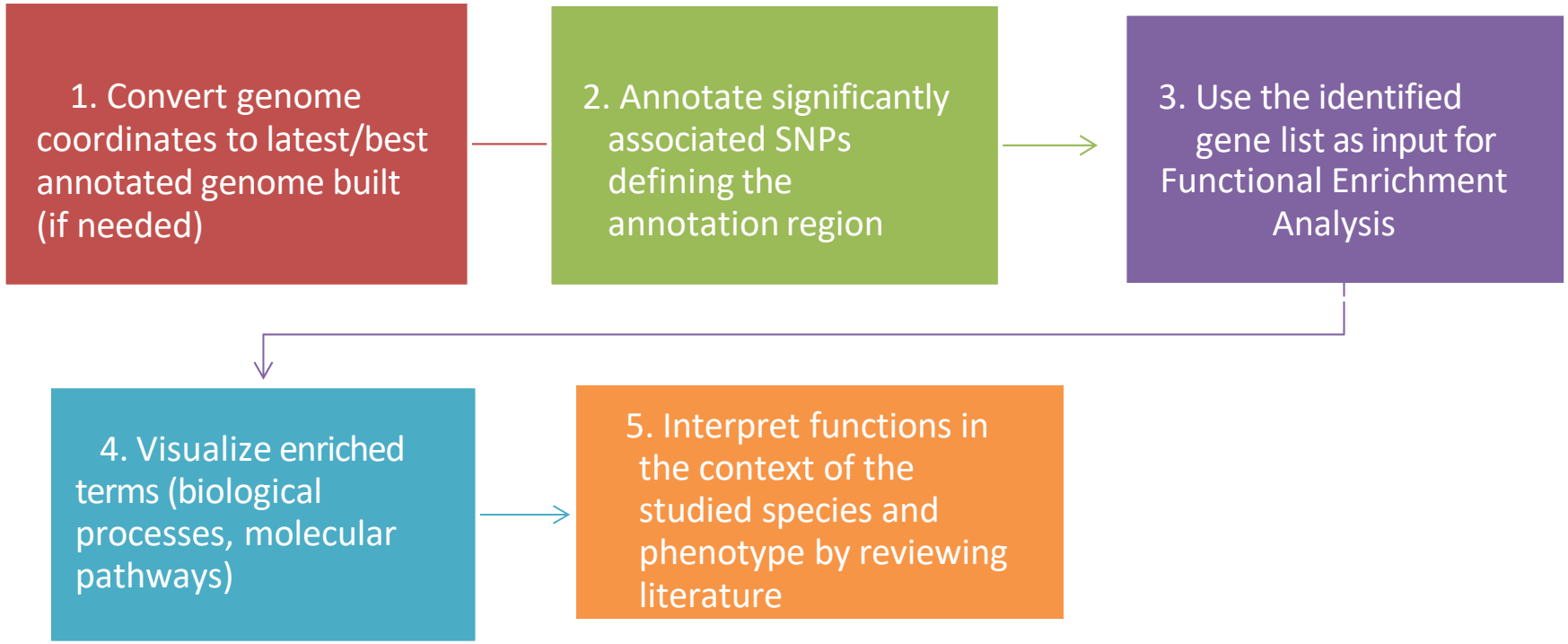
https://genome.ucsc.edu/cgi-bin/hgTracks?db=oviAri4&lastVirtModeType=default&lastVirtModeExtraState=&virtModeType=default&virtMode=0&nonVirtPosition=&position=chr3%3A130065882%2D160065881&hgsid=3912947233_JH7X2OATijgk95vyEagS6lQe7dUL

Ensembl Genome Browser

https://www.ensembl.org/Ovis_aries/Location/View?r=3:145487390-145490781;db=core

However, investigating markers individually is not practical in case of hundreds or thousands significantly associated SNPs

Workflow Summary



1. LiftOver Genome Coordinates between different assemblies

widely used options to convert genome coordinates in another assembly

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Tool:Converting Genome Coordinates From One Genome Version To Another (Ucsc Liftover, Ncbi Remap, Ensembl Api)

195

Some recent posts reminded me that it might be useful for us to review the options for converting between genome coordinate systems.

This comes up in several contexts. Probably the most common is that you have some coordinates for a particular version of a reference genome and you want to determine the corresponding coordinates on a different version of the reference genome for that species. For example, you have a bed file with exon coordinates for human build GRC37 (hg19) and wish to update to GRCh38. By the way, for a nice summary of genome versions and their release names refer to the Assembly Releases and Versions FAQ

Or perhaps you have coordinates of a gene and wish to determine the corresponding coordinates in another species. For example, you have coordinates of a gene in human GRCh38 and wish to determine corresponding coordinates in mouse mm10.

13.1 years ago
Malachi Griffith 20k

Finally you may wish to convert coordinates between coordinate systems within a single assembly. For example, you have the coordinates of a series of exons and you want to determine the position of these exons with respect to the transcript, gene, contig, or entire chromosome.

There are now several well known tools that can help you with these kinds of tasks:

1. **UCSC liftOver**. This tool is available through a simple web interface or it can be downloaded as a standalone executable. To use the executable you will also need to download the appropriate chain file. Each chain file describes conversions between a pair of genome assemblies. Liftover can be used through Galaxy as well. There is a python implementation of liftover called pyliftover that does conversion of point coordinates only.

2. **NCBI Remap**. This tool is conceptually similar to liftOver in that in manages conversions between a pair of genome assemblies but it uses different methods to achieve these mappings. It is also available through a simple web interface or you can use the API for NCBI Remap.

3. **The Ensembl API**. The final example I described above (converting between coordinate systems within a single genome assembly) can be accomplished with the Ensembl core API. Many examples are provided within the installation, overview, tutorial and documentation sections of the Ensembl API project. In particular, refer to these sections of the tutorial: 'Coordinates', 'Coordinate systems', 'Transform', and 'Transfer'.

4. **Assembly Converter**. Ensembl also offers their own simple web interface for coordinate conversions called the Assembly Converter.

5. **Bioconductor rtracklayer** package. For R users, Bioconductor has an implementation of UCSC liftOver in the rtracklayer package. To see documentation on how to use it, open an R session and run the following commands.

```
source("http://bioconductor.org/biocLite.R")
biocLite("rtracklayer")
library(rtracklayer)
?liftOver
```

6. **CrossMap**. A standalone open source program for convenient conversion of genome coordinates (or annotation files) between different assemblies. It supports most commonly used file formats including SAM/BAM, Wiggle/BigWig, BED, GFF/GTF, VCF. CrossMap is designed to liftover genome coordinates between assemblies. It's not a program for aligning sequences to reference genome. Not recommended for converting genome coordinates between species.

7. Flo. A liftover pipeline for different reference genome builds of the same species. It describes the process as follows: "align the new assembly with the old one, process the alignment data to define how a coordinate or coordinate range on the old assembly should be transformed to the new assembly, transform the coordinates."

8. Picard Liftover VCF. Lifts over a VCF file from one reference build to another. This tool adjusts the coordinates of variants within a VCF file to match a new reference. The tool is based on the UCSC liftOver and uses a UCSC chain file to guide its operation.

https://www.biostars.org/p/65558/#391077

CrossMap
Installed in python virtual environment

1. LiftOver Genome Coordinates between different assemblies

Input file: BED, VCF, GFF

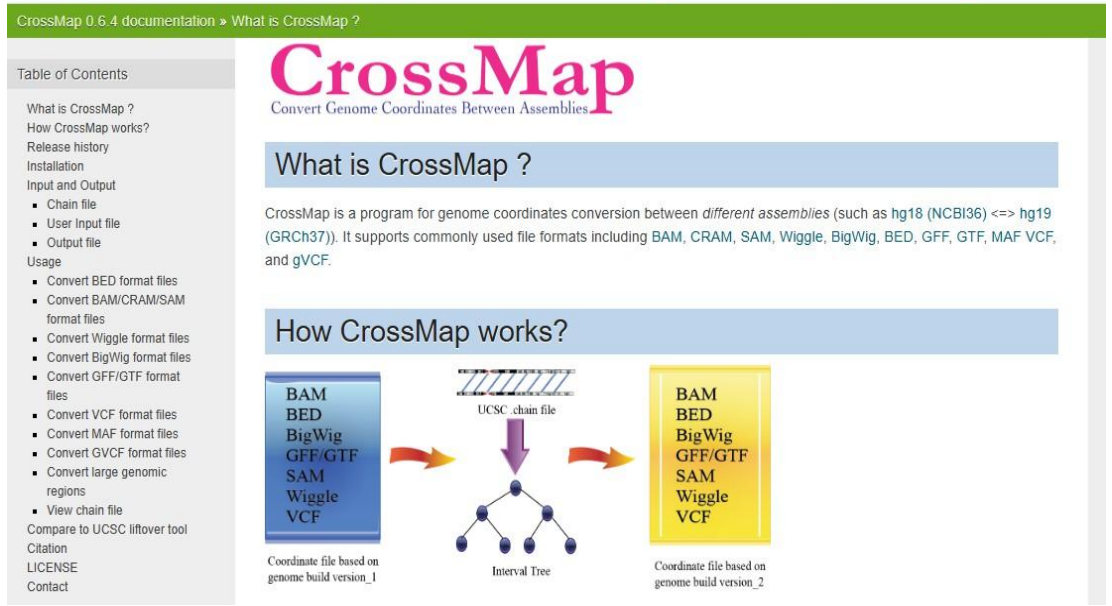
File format →

chr3	189803626	189803626	OAR3_204624914.1
chr3	145065882	145065882	OAR3_155103999.1
chr3	131220126	131220126	OAR3_140271376.1
chr3	181785624	181785624	OAR3_195793605_X.1

Chain file

text file encoding alignments between different assemblies in segment blocks

contiguous aligned segments defining also the gaps in both target and query genomes



More info: https://crossmap.sourceforge.net/

Chain files for sheep genome provided by UCSC Genomics Institute

↓

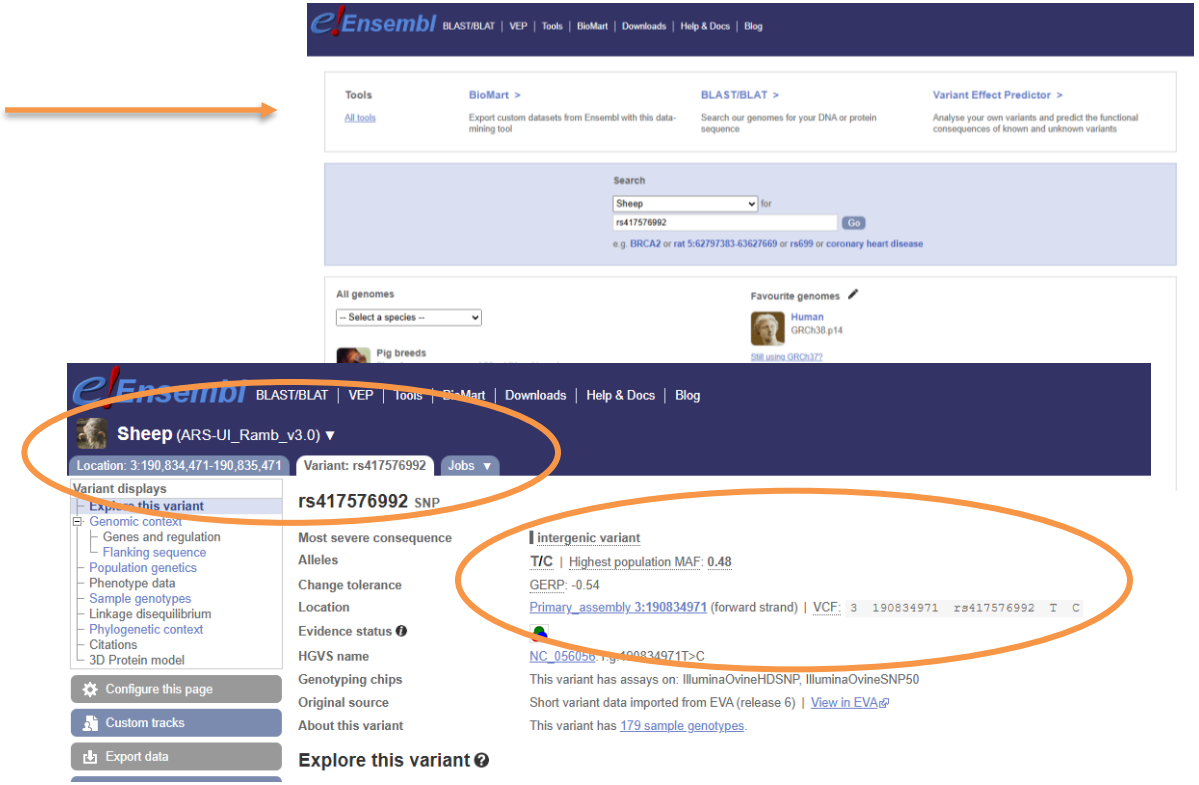
https://groups.google.com/a/soe.ucsc.edu/g/genome/c/2BSf_4qAyZY

1. LiftOver Genome Coordinates between different assemblies

How do we trace the genome built that our data correspond to?

1. Use a variant ID (rs ID) in a genome browser that is based on a specific genome built of the studied species

2. If the position does not match the map file, use online liftover tool to determine the assembly of your data



Our data correspond to the *Ovis aries* genome assembly Oar_v4.0 → Texel breed - 2015

Conversion of genome coordinates using ARS-UI_Ramb_v2.0 in-depth functional annotation of the sheep genome (https://academic.oup.com/gigascience/article/doi/10.1093/gigascience/giab096/6521876) → Rambouillet breed - 2022

2. Annotate significantly associated SNPs

Ensembl VEP (Variant Effect Predictor) – Variant Annotation

1 Input file: a list with the identified variants

VEP default input →

chromosome
start position
end position
allele
strand

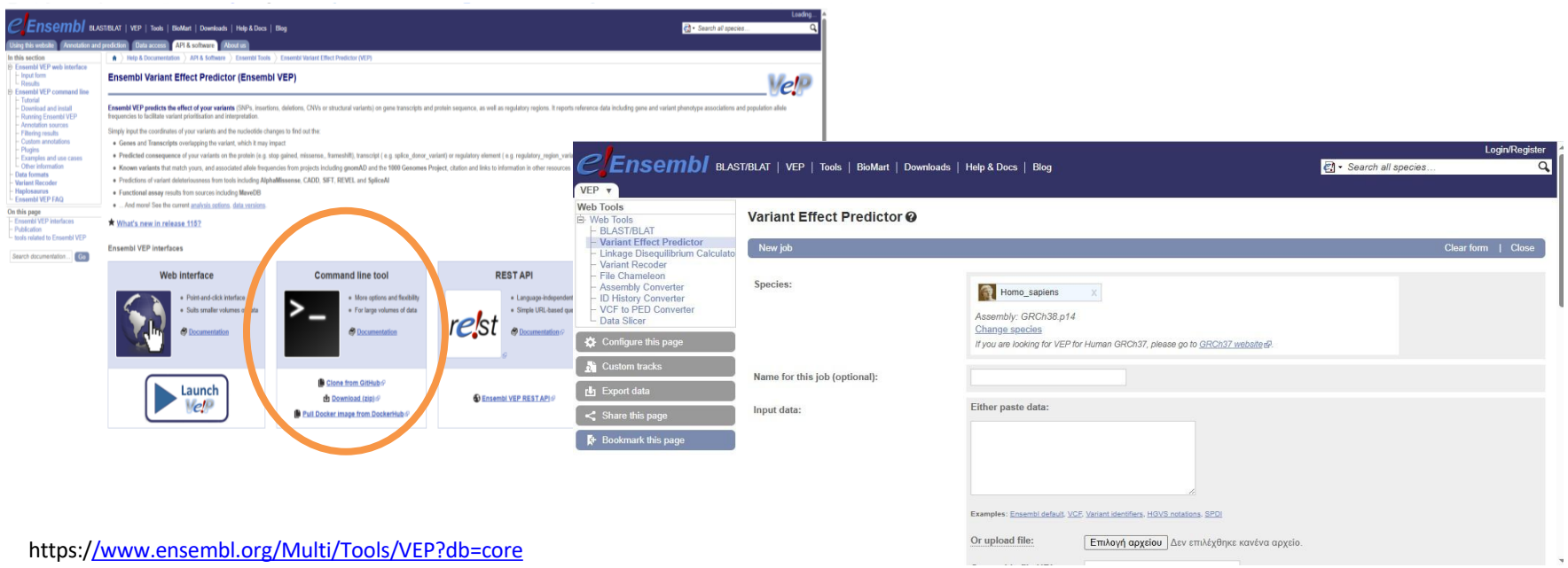
https://www.ensembl.org/info/docs/tools/vep/vep_formats.html#default

Daily grazing duration

3 190036206 190036206 A/G
3 145252109 145252109 A/G
3 131380348 131380348 G/A
3 181785624 181785624 G/A
7 62298200 62298200 A/C
7 4217464 4217464 A/G
11 26623188 26623188 A/C
11 29347981 29347981 A/G
11 18248852 18248852 A/G
11 49829927 49829927 C/A
14 39858010 39858010 A/G
14 39795104 39795104 A/G
19 10373068 10373068 G/A

Altitude difference

15 71858906 71858906 G/A
15 63232034 63232034 G/A
15 3499482 3499482 G/A
7 7334440 7334440 C/A
3 18569142 18569142 A/C



https://www.ensembl.org/Multi/Tools/VEP?db=core

2. Annotate significantly associated SNPs

Ensembl VEP (Variant Effect Predictor) – Variant Annotation

2

-Cache file = Ensembl annotation source: **refers to a specific genome built**

- File with all transcripts, regulatory features and variant data for a species
- **It should be compatible to the vep version installed**
  https://www.ensembl.org/info/docs/tools/vep/script/vep_cache.html#cache

Examples and use cases
Other information

On this page
Caches
GFF/GTF files
FASTA files
Databases
Cache - technical information

Search documentation… Go

- Cache - a downloadable file containing all transcript models, regulatory features and variant data for a species
- GFF or GTF - use transcript models defined in a tabix-indexed GFF or GTF file
  - Requires a FASTA file in --offline mode or if the desired species or assembly is not part of the Ensembl species list.
- Database - connect to a MySQL database server hosting Ensembl databases

Data from VCF, BED and bigWig files can also be incorporated by Ensembl VEP's Custom annotation feature.

Using a cache is the most efficient way to use Ensembl VEP; we would encourage you to use a cache wherever possible. Caches are easy to download and set up using the installer. Follow the tutorial for a simple guide.

Caches

Using a cache (--cache) is the fastest and most efficient way to use Ensembl VEP, as in most cases only a single initial network connection is made and most data is read from local disk. Use offline mode to eliminate all network connections for speed and/or privacy.

Cache version

We strongly recommend that you download/use the cache version which corresponds to your Ensembl VEP installation, i.e. cache version **115** should be used with the Ensembl VEP tool version **115**.

This is mainly due to the fact that the cache (data content and structure) is generated every Ensembl release, regarding the data and API updates for this release, therefore the cache data format might differ between versions (and be incompatible with a newer version of the Ensembl VEP tool).

Downloading caches

Cache files are created for every species for each Ensembl release. They can be automatically downloaded and configured using INSTALL.pl.

If interested in RefSeq transcripts you may download an alternate cache file (e.g. homo_sapiens_refseq), or a merged file of RefSeq and Ensembl transcripts (eg homo_sapiens_merged); remember to specify --refseq or --merged when running Ensembl VEP to use the relevant cache. See documentation for full details.

Manually downloading caches

It is also simple to download and set up caches without using the installer. By default, Ensembl VEP searches for caches in $HOME/.vep; to use a different directory when running Ensembl VEP, use --dir_cache.

Indexed cache (https://ftp.ensembl.org/pub/release-115/variation/indexed_vep_cache/)

Essential for human and other species with large sets of variant data - requires Bio::DB::HTS (setup by INSTALL.pl) or tabix, e.g.:

```
cd $HOME/.vep
curl -O https://ftp.ensembl.org/pub/release-115/variation/indexed_vep_cache/homo_sapiens_vep_115_GRCh38.tar.gz
tar xzf homo_sapiens_vep_115_GRCh38.tar.gz
```

FTP directories with indexed cache data:

Ensembl:	Vertebrates
Ensembl Genomes:	Bacteria | Fungi | Metazoa | Plants | Protists

Please use the web portal with 1 core per session to install and run VEP!

2. Annotate significantly associated SNPs

Example Output – VEP

Gene Ensembl ID
Consequence
cDNA position
Distance from SNP

```
## Codons : Reference and variant codon sequence
## Existing_variation : Identifier(s) of co-located known variants
## Extra column keys:
## IMPACT : Subjective impact classification of consequence type
## DISTANCE : Shortest distance from variant to transcript
## STRAND : Strand of the feature (1/-1)
## FLAGS : Transcript quality flags
## VEP command-line: vep --cache --database 0 --distance 500000 --input_file input_vep.Ensembl --output_file output_vep.Ensembl --species ovis_aries
#Uploaded_variation     Location        Allele  Gene    Feature Feature_type    Consequence     cDNA_position   CDS_position    Protein_position        Amino_acids     Codons  Existing_variation      Extra
3_190834971_A/G 3:190834971     G       ENSOARG00020039732      ENSOART00020005659      Transcript      upstream_gene_variant   -       -       -       -       -       -       IMPACT=MODIFIER;DISTANCE=409525;STRAND=-1
3_190834971_A/G 3:190834971     G       ENSOARG00020003767      ENSOART00020006508      Transcript      downstream_gene_variant -       -       -       -       -       -       IMPACT=MODIFIER;DISTANCE=319939;STRAND=1
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3_190834971_A/G 3:190834971     G       ENSOARG00020003767      ENSOART00020006708      Transcript      downstream_gene_variant -       -       -       -       -       -       IMPACT=MODIFIER;DISTANCE=320098;STRAND=1
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3_131527909_G/A 3:131527909     A       ENSOARG00020014571      ENSOART00020022295      Transcript      downstream_gene_variant -       -       -       -       -       -       IMPACT=MODIFIER;DISTANCE=61279;STRAND=1
3_131527909_G/A 3:131527909     A       ENSOARG00020014635      ENSOART00020023247      Transcript      upstream_gene_variant   -       -       -       -       -       -       IMPACT=MODIFIER;DISTANCE=88214;STRAND=1
3_131527909_G/A 3:131527909     A       ENSOARG00020014635      ENSOART00020023352      Transcript      upstream_gene_variant                                                   IMPACT=MODIFIER;DISTANCE=88214;STRAND=1
```

3. Functional Annotation

Process of determining gene functions, pathways associated with specific sequences/genes

Gene Ontology (GO) = the widest annotation source of gene products

standardized system to describe gene functions and their products across species in a consistent way with structured vocabulary

GO Has Three Main Categories-Sources:

- **Biological Process (BP)**
  What the gene product does in a wide biological context.
  e.g., cell division, immune response, signal transduction.
- **Molecular Function (MF)**
  The basic activity of the gene product at the molecular level. e.g., ATP binding, enzyme activity, receptor binding.
- **Cellular Component (CC)**
  Where in the cell the gene product is active.
  e.g., nucleus, mitochondrion, plasma membrane.

Types of functional annotation

Code	Meaning
EXP	Inferred from Experiment
IDA	Inferred from Direct Assay
IEA	Inferred from Electronic Annotation
ISS	Inferred from Sequence or Structural Similarity
TAS	Traceable Author Statement
IMP	Inferred from Mutant Phenotype

3. Functional Enrichment Annotation

definition

A statistical test to determine whether certain functions/pathways are over-represented in a defined set of genes

steps

- Gene list as an input
- Select a reference genome including all annotated genes
- Test each annotation term (GO term) for enrichment
- Correct for multiple testing (e.g., Bonferroni, FDR)

3. Functional Enrichment Analysis

g:Profiler2 (R package) - toolset for functional enrichment analysis and visualization

https://pmc.ncbi.nlm.nih.gov/articles/PMC7859841/pdf/f1000research-9-29655.pdf

- Utilizes data from multiple reliable databases GO, KEGG, Reactome etc. for functional annotation across various species
- Input: gene list obtained from VEP tool
- Format: Ensembl gene IDs (preferred), Gene symbols

Main functions
gost(): functional profiling
gostplot(): plot the enrichment results
gconvert(): naming conventions between genes, proteins etc across databases and species
gorth(): convert a gene list from a source organism to a target one

https://biit.cs.ut.ee/gprofiler/page/r

3. Functional Enrichment Analysis

Utilization of gprofiler2 (R package)

Install and load gprofiler2 library in RStudio

1. Example input (single query)

```
</> R
gostres <- gost(query = c("X:1000:1000000", "rs17396340", "GO:0005005", "ENSG00000156103", "NLRP1"), organism = "hsapiens")
```

2. Functional annotation from GWAS-derived gene lists (VEP tool results for two SNPs)

Create a named list object in R including two gene lists (Ensembl gene IDs) that correspond to two SNP markers

```
</> R
gene_lists <- list(…)
```

Run functional enrichment analysis - find the argument for multiple gene lists (search in R documentation), optional parameters for significance, threshold, correction)

```
</> R
gostres <- gost(query = gene_lists, organism = "oarambouillet", …, highlight=TRUE)
```

3. Functional Enrichment Vizualization

Utilization of gprofiler2 (R package)

Explore functional enrichment results through an interactive Manhattan plot

```
</> R

gostplot(gost_output, capped=TRUE, interactive=TRUE)
```

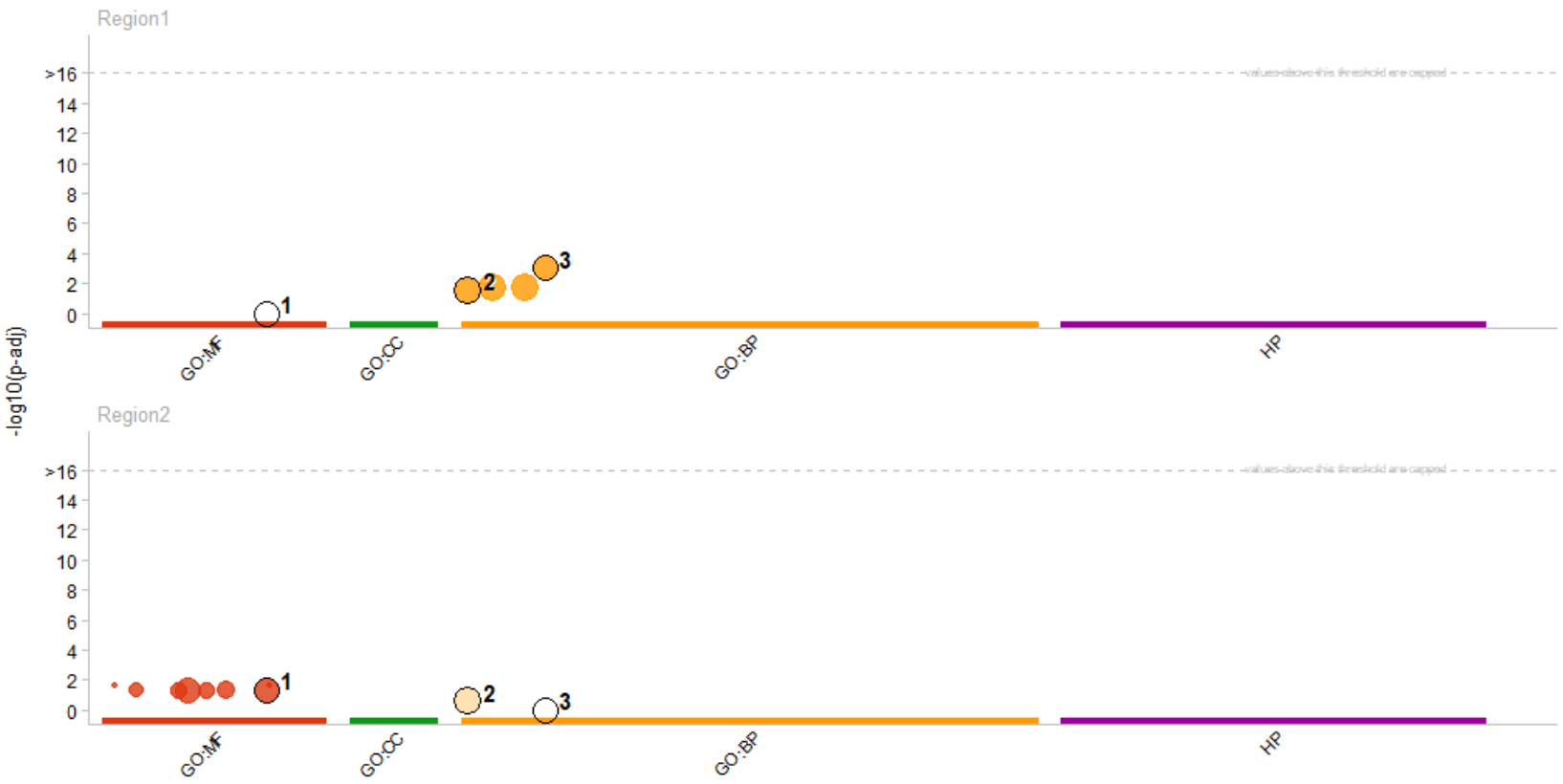
Visualize enrichment results selecting specific GO terms from both annotated gene lists using the interactive Manhattan plot

```
</> R

publish_gostplot(gostplot_output, highlight_terms= ...)
```

Functional Enrichment Results

Manhattan plot of enriched GO terms using 2 gene lists (chr3) as an input



id	source	term_id	term_name	term_size		p_value Region1		p_value Region2
1	GO:MF	GO:0051020	GTPase binding	201		1.0e+00		4.6e-02
2	GO:BP	GO:0001501	skeletal system development	398		3.0e-02		2.8e-01
3	GO:BP	GO:0009952	anterior/posterior pattern specification	171		9.9e-04		1.0e+00

g:Profiler (biit.cs.ut.ee/gprofiler)

Functional Enrichment - Vizualization

Usage of alternative R packages for visualization purposes using the output from gprofiler2 gost()

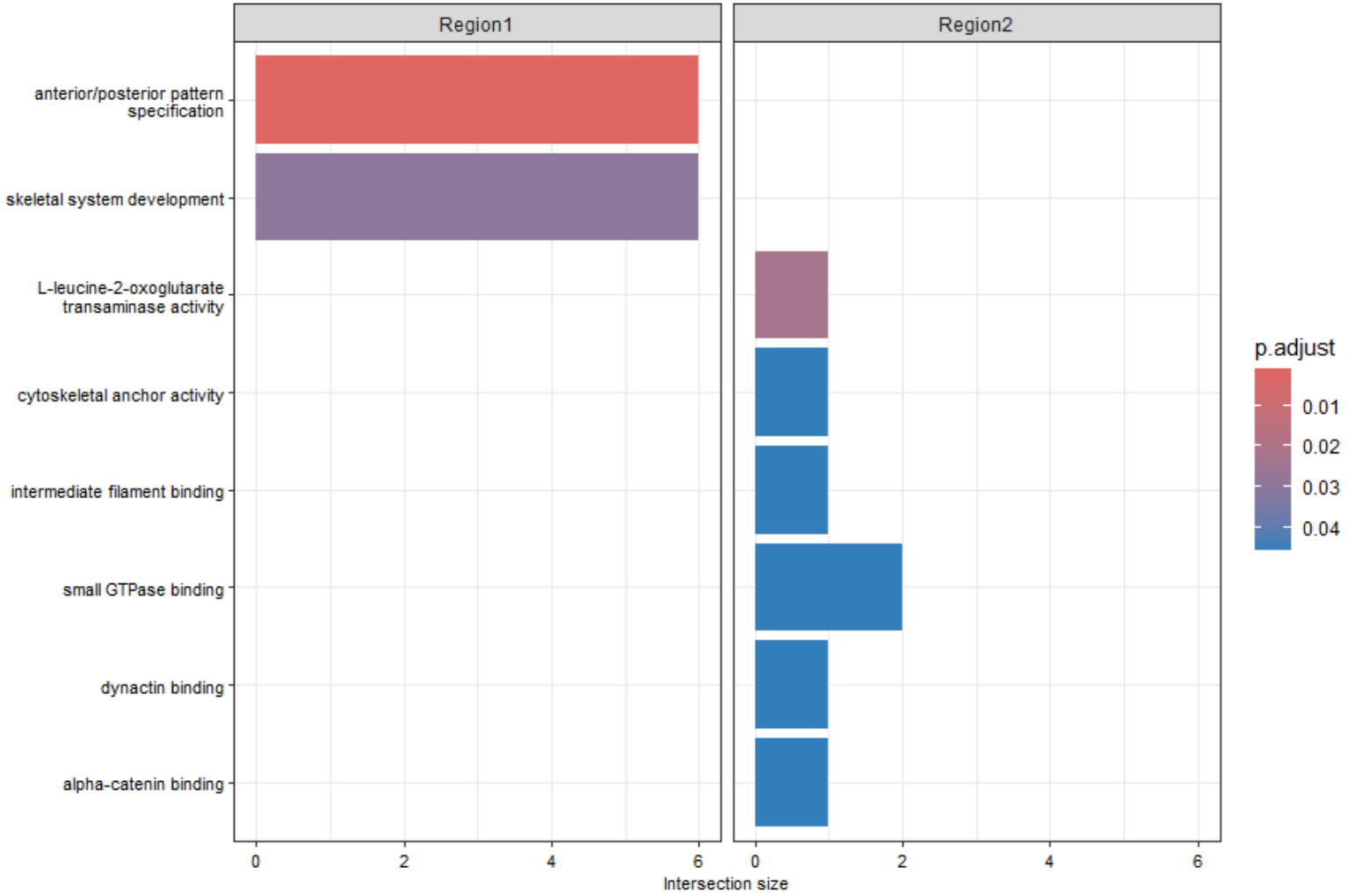
clusterProfiler
enrichplot
DOSE
ggplot2

Specific columns are required from the data frame produced by gost(), such as

query,
source,
term_id,
term_name,
p_value,
query_size,
intersection_size,
term_size,
effective_domain_size,
intersection

Functional Enrichment Results

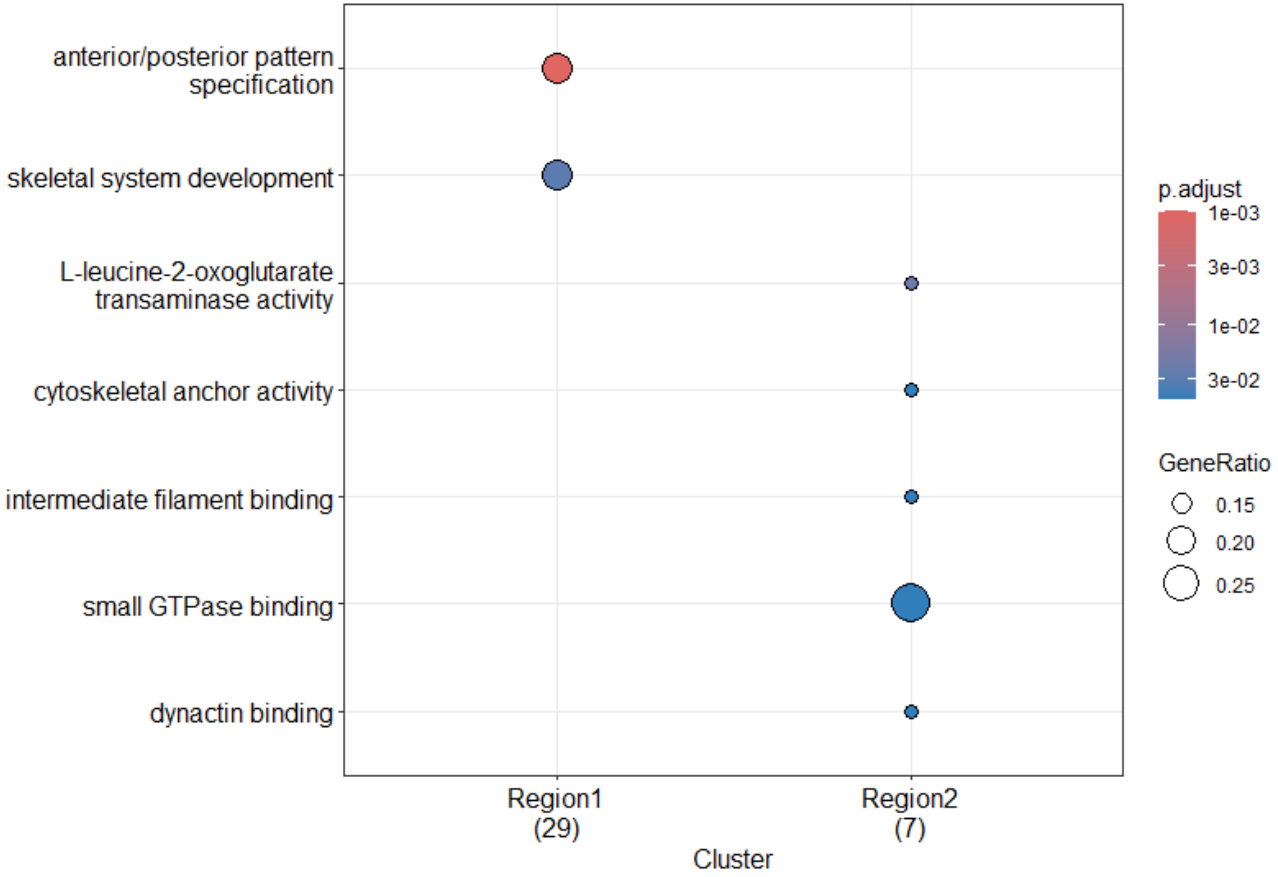
Bar plot of enriched GO terms showing number of candidate genes in each biological process detected



barplot()

Functional Enrichment Results

Dot plot of enriched GO terms showing number of candidate genes in each biological process detected



enrichplot()

Use the link to access the material for annotation workflow

https://github.com/valentinatsar/annotation/tree/main